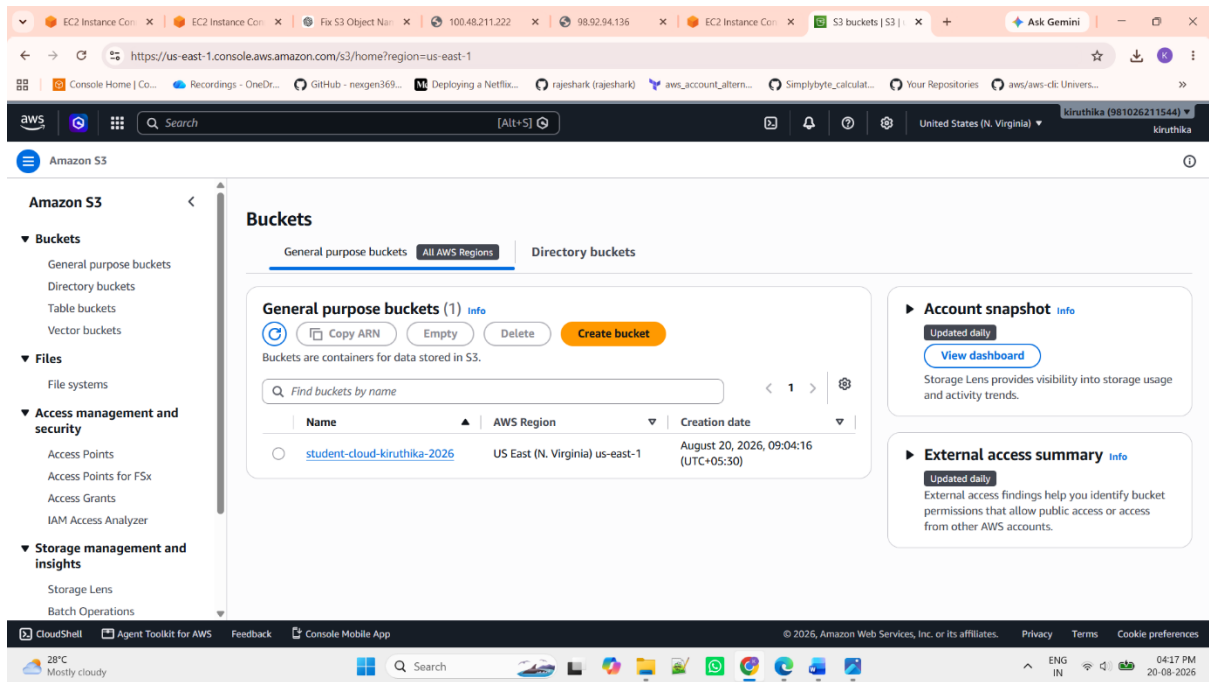
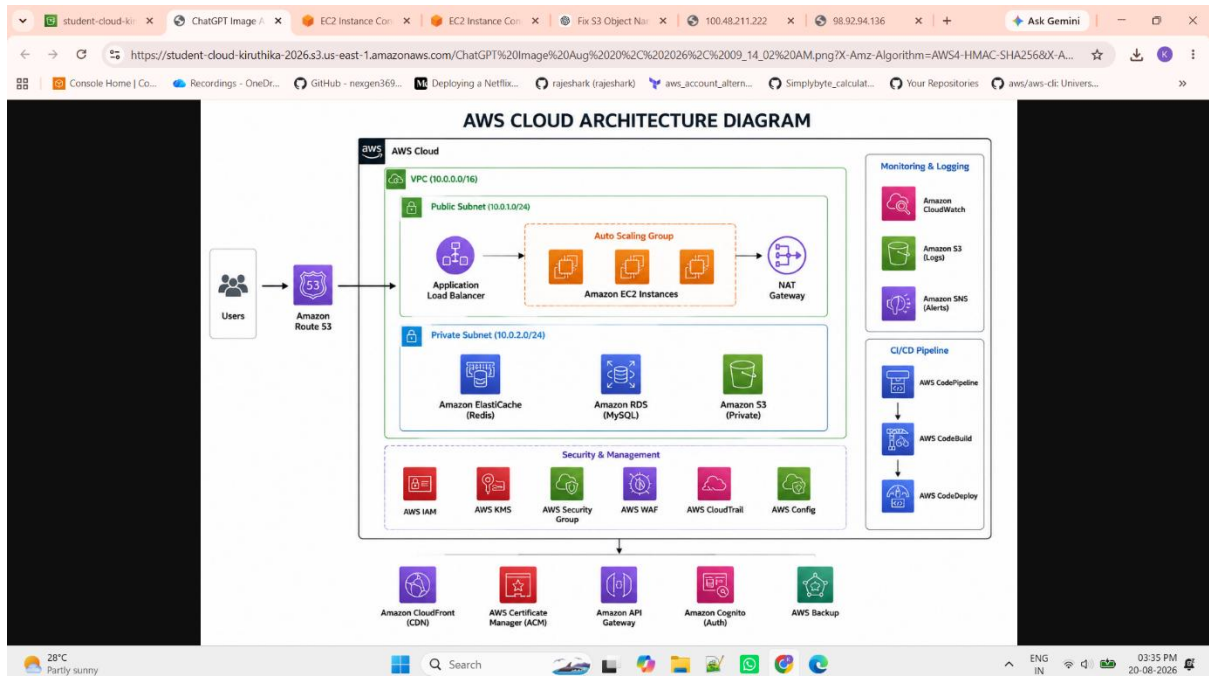


Task1:S3

S3 bucket created:



cloud.jpg:



Student-guide.pdf:

AWS DevOps Student Guide

A practical beginner-to-intermediate reference for AWS infrastructure, security, databases, networking, monitoring, and DevOps workflows.

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notes.txt:

AWS DEVOPS STUDENT NOTES

1. **AWS FUNDAMENTALS** AWS is a cloud platform providing compute, storage, networking, databases, security, monitoring, and managed services.

Region: A geographic AWS location.

Availability Zone: An isolated location inside a Region. Deploying across multiple AZs improves availability.

Shared Responsibility: AWS secures the cloud infrastructure. Customers secure their data, identities, applications, and configurations.

2. **IAM AND SECURITY** IAM controls access to AWS resources.

Key components:

- User: identity for a person or application.
- Group: collection of IAM users.
- Role: identity that can be assumed and provides temporary credentials.
- Policy: JSON document defining permissions.
- MFA: additional authentication factor.
- Least privilege: provide only required permissions.

3. **S3** S3 is object storage. Objects are stored in buckets.

Important concepts:

- Bucket: container for objects.
- Object: file plus metadata.
- Versioning: keeps multiple object versions.
- Lifecycle: automatically transitions or deletes objects.
- Block Public Access: prevents unintended public exposure.

Private S3: Keep Block Public Access enabled and grant access only to authorized identities/services.

Pre-signed URL: Provides temporary access to a private object without requiring the end user to have AWS credentials.

4. **EC2** EC2 provides virtual servers.

Important concepts:

- AMI: image used to launch an instance.
- Instance type: determines CPU, memory, networking, etc.
- EBS: persistent block storage.
- Security Group: stateful virtual firewall.
- User Data: startup/bootstrap script.
- Key pair: commonly used for SSH authentication.

5. **VPC AND NETWORKING** VPC is a logically isolated AWS network.

Components:

- CIDR: defines IP address range.
- Subnet: smaller IP range

Private/Block Public Access:

The screenshot shows the AWS Management Console for the 'student-cloud-kiruthika-2026' S3 bucket. The 'Permissions' tab is selected, and the 'Block public access (bucket settings)' section is expanded. It shows that 'Block all public access' is turned 'On'. The console interface includes a left-hand navigation menu with options like 'Buckets', 'Files', 'Access management and security', and 'Storage management and insights'. The top bar shows the user 'kiruthika (981026211544)' and the region 'United States (N. Virginia)'.

Task2: IAM

StudentCloudEC2Role:

The screenshot shows the AWS Management Console for the 'StudentCloudEC2Role' IAM role. The 'Summary' tab is selected, displaying details such as 'Creation date' (August 20, 2026), 'ARN' (arn:aws:iam::981026211544:role/StudentCloudEC2Role), and 'Instance profile ARN' (arn:aws:iam::981026211544:instance-profile/StudentCloudEC2Role). The 'Permissions policies' section shows one policy attached, 's3-policy', which is a 'Customer managed' policy. The console interface includes a left-hand navigation menu with options like 'Dashboard', 'Access Management', 'Policies', 'IAM users', 'IAM user groups', 'Account settings', 'Root access management', 'Temporary delegation requests', 'Access reports', 'Resource analysis', 'Unused access', and 'Analyzer settings'. The top bar shows the user 'kiruthika (981026211544)' and the region 'Global'.

Custom policy:

The screenshot shows the AWS IAM console interface. The left sidebar contains navigation links for Identity and Access Management (IAM), Access Management, and Access reports. The main content area displays the details for a policy named 's3-policy'. The policy details section shows the type as 'Customer managed', creation time as 'August 20, 2026, 09:28 (UTC+05:30)', edited time as 'August 20, 2026, 13:32 (UTC+05:30)', and ARN as 'arn:aws:iam::981026211544:policy/s3-policy'. Below this, the 'Permissions' tab is selected, showing a table of permissions defined in the policy. The table has columns for Service, Access level, Resource, and Request condition. The only permission listed is for the 'S3' service with 'Limited: List, Read' access level, 'Multiple' resource, and 'None' request condition. There are buttons for 'Edit', 'Summary', and 'JSON' for the permissions.

aws sts get-caller-identity:

```
15 history
[ec2-user@ip-172-31-13-234 ~]$ aws sts get-caller-identity
{
  "UserId": "AROA6I2NYA3MBNMMKKKBF:i-0a07daa5308b61e77",
  "Account": "981026211544",
  "Arn": "arn:aws:sts::981026211544:assumed-role/StudentCloudEC2Role/i-0a07daa5308b61e77"
}
```

aws s3 ls:

```
    "Account": "981026211544",
    "Arn": "arn:aws:sts::981026211544:assumed-role/StudentCloudEC2Role/i-0a07daa5308b61e77"
}
[ec2-user@ip-172-31-13-234 ~]$ aws s3 ls s3://student-cloud-kiruthika-2026
2026-08-20 03:47:44  1285497 ChatGPT Image Aug 20, 2026, 09_14_02 AM.png
2026-08-20 03:47:44      6280 notes.txt
2026-08-20 03:47:45   41540 student-guide.pdf
```

aws s3 cp:

```

[ec2-user@ip-172-31-13-234 ~]$ aws s3 ls s3://student-cloud-kiruthika-2026
2026-08-20 03:47:47  1285497 ChatGPT Image Aug 20, 2026, 09_14_02 AM.png
2026-08-20 03:47:44      6280 notes.txt
2026-08-20 03:47:45   41540 student-guide.pdf
[ec2-user@ip-172-31-13-234 ~]$ aws s3 cp "s3://student-cloud-kiruthika-2026/ChatGPT Image Aug 20, 2026, 09_14_02 AM.png" .
download: s3://student-cloud-kiruthika-2026/ChatGPT Image Aug 20, 2026, 09_14_02 AM.png to ./ChatGPT Image Aug 20, 2026, 09_14_02 AM.png
```

aws s3 rm cloud.jpg → AccessDenied:

```

[ec2-user@ip-172-31-13-234 ~]$ aws s3 cp "s3://student-cloud-kiruthika-2026/ChatGPT Image Aug 20, 2026, 09_14_02 AM.png" .
download: s3://student-cloud-kiruthika-2026/ChatGPT Image Aug 20, 2026, 09_14_02 AM.png to ./ChatGPT Image Aug 20, 2026, 09_14_02 AM.png
[ec2-user@ip-172-31-13-234 ~]$ aws s3 rm s3://student-cloud-kiruthika-2026/student-guide.pdf
delete failed: s3://student-cloud-kiruthika-2026/student-guide.pdf An error occurred (AccessDenied) when calling the DeleteObject operation: User: arn:aws:sts::981026211544:assumed-role/StudentCloudEC2Role/i-0a07daa5308b61e77 is not authorized to perform: s3:DeleteObject on resource: "arn:aws:s3:::student-cloud-kiruthika-2026/student-guide.pdf" because no identity-based policy allows the s3:DeleteObject action
[ec2-user@ip-172-31-13-234 ~]$
```

Task3:EC2

EC2 instance:

The screenshot shows the AWS Management Console for the 'us-east-1' region. The 'Instances' page is active, displaying a list of EC2 instances. The instance 'student-cloud-ec2' with ID 'i-0a07daa5308b61e77' is shown in a 'Running' state. The console also displays various status checks, service health, and health events.

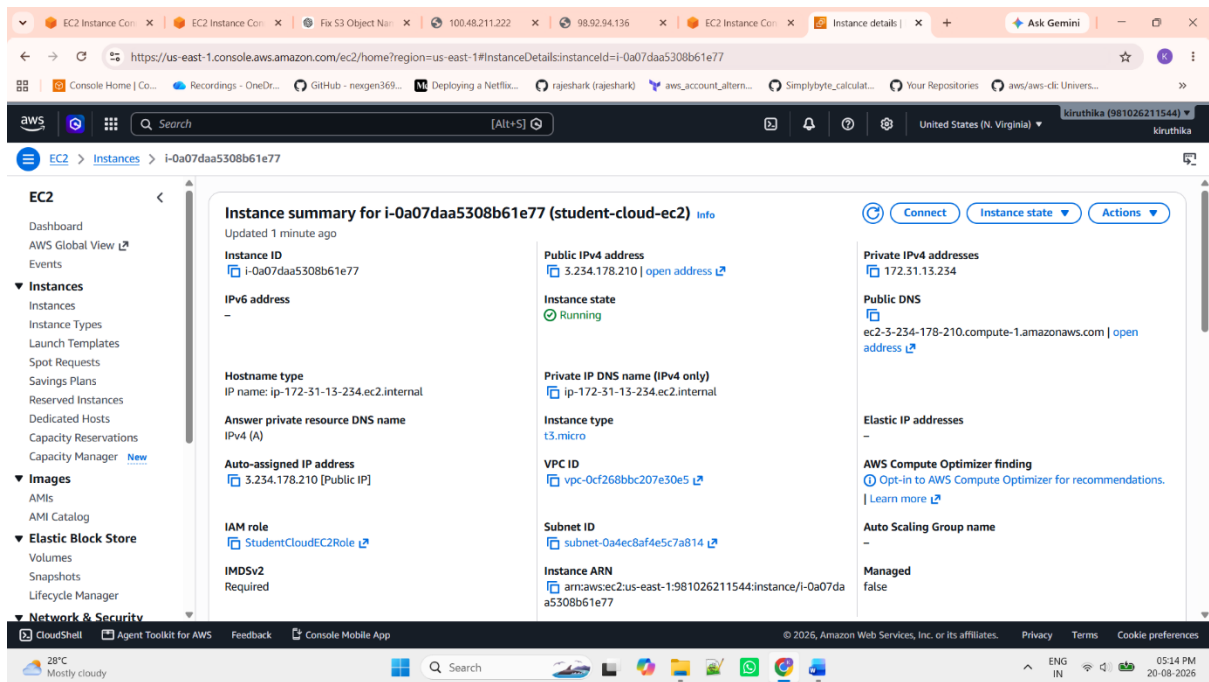
Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DI
student-cloud-ec2	i-0a07daa5308b61e77	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-3-234-17i

Amazon Linux:

The screenshot shows the 'Instance details' page for the EC2 instance 'student-cloud-ec2'. The page provides comprehensive information about the instance, including its AMI, launch time, and various configuration settings.

Details	Status and alarms	Monitoring	Security	Networking	Storage	Tags
Instance details AMI ID : ami-0332d564d76dbd8d6 AMI name : al2023-ami-2023.12.20260817.0-kernel-6.18-x86_64 Stop protection : Disabled Instance reboot migration : Default (On) Stop-hibernate behavior : Disabled State transition reason : - State transition message : - Owner : 981026211544	Monitoring : disabled Allowed image : - Launch time : Thu Aug 20 2026 13:13:08 GMT+0530 (India Standard Time) (about 4 hours) Instance auto-recovery : Default AMI Launch index : 0 Credit specification : unlimited Usage operation : RunInstances Enclaves Support : Disabled	Platform details : Linux/UNIX Termination protection : Disabled AMI location : amazon/al2023-ami-2023.12.20260817.0-kernel-6.18-x86_64 Lifecycle : normal Key pair assigned at launch : student-cloud-key Kernel ID : - RAM disk ID : - Boot mode : uefi-preferred				

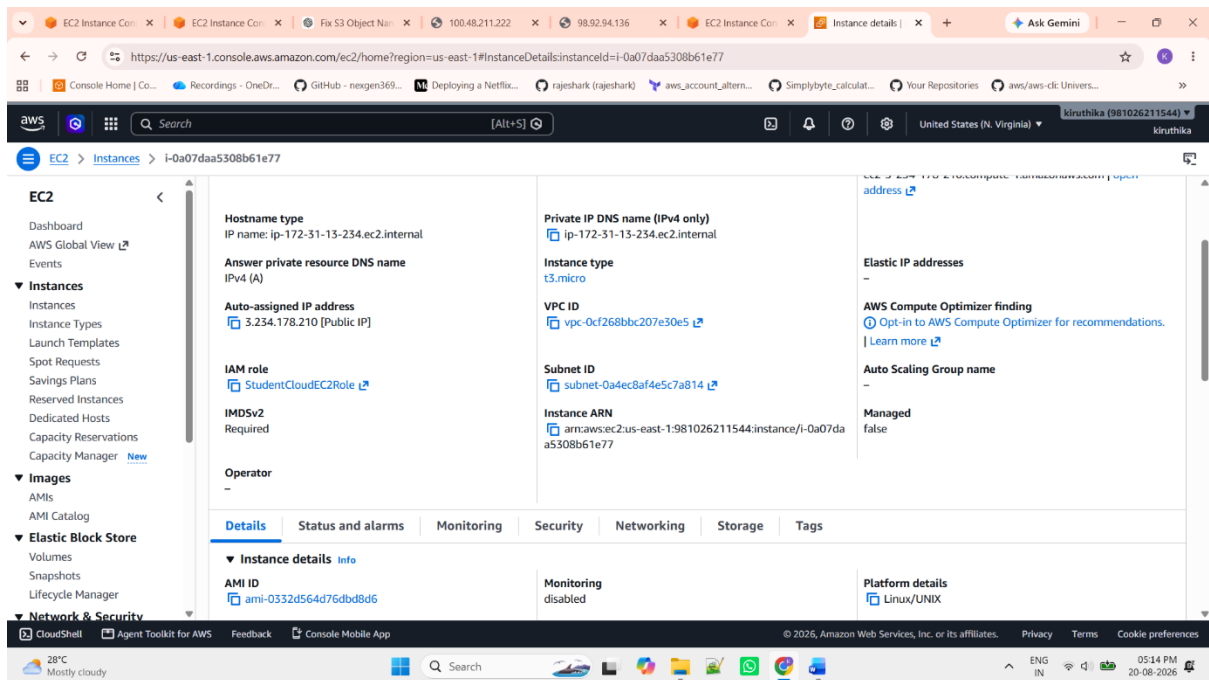
Private IP and Public IP:



The screenshot displays the AWS Management Console for an EC2 instance named **i-0a07daa5308b61e77** (student-cloud-ec2). The instance is in the **Running** state. The console shows various details including the instance ID, public and private IP addresses, hostname, DNS names, VPC ID, subnet ID, and instance type (**t3.micro**). The instance is running on the **Amazon Linux 2** AMI. The console also shows the instance's location (us-east-1) and the region (United States (N. Virginia)).

Field	Value
Instance ID	i-0a07daa5308b61e77
Public IPv4 address	3.234.178.210
Private IPv4 addresses	172.31.13.234
Public DNS	ec2-3-234-178-210.compute-1.amazonaws.com
Private IP DNS name (IPv4 only)	ip-172-31-13-234.ec2.internal
Instance type	t3.micro
VPC ID	vpc-0cf268bbc207e30e5
Subnet ID	subnet-0a4ec8af4e5c7a814
Instance ARN	arn:aws:ec2:us-east-1:981026211544:instance/i-0a07daa5308b61e77
IAM role	StudentCloudEC2Role
IMDSv2	Required
Auto Scaling Group name	-
Managed	false

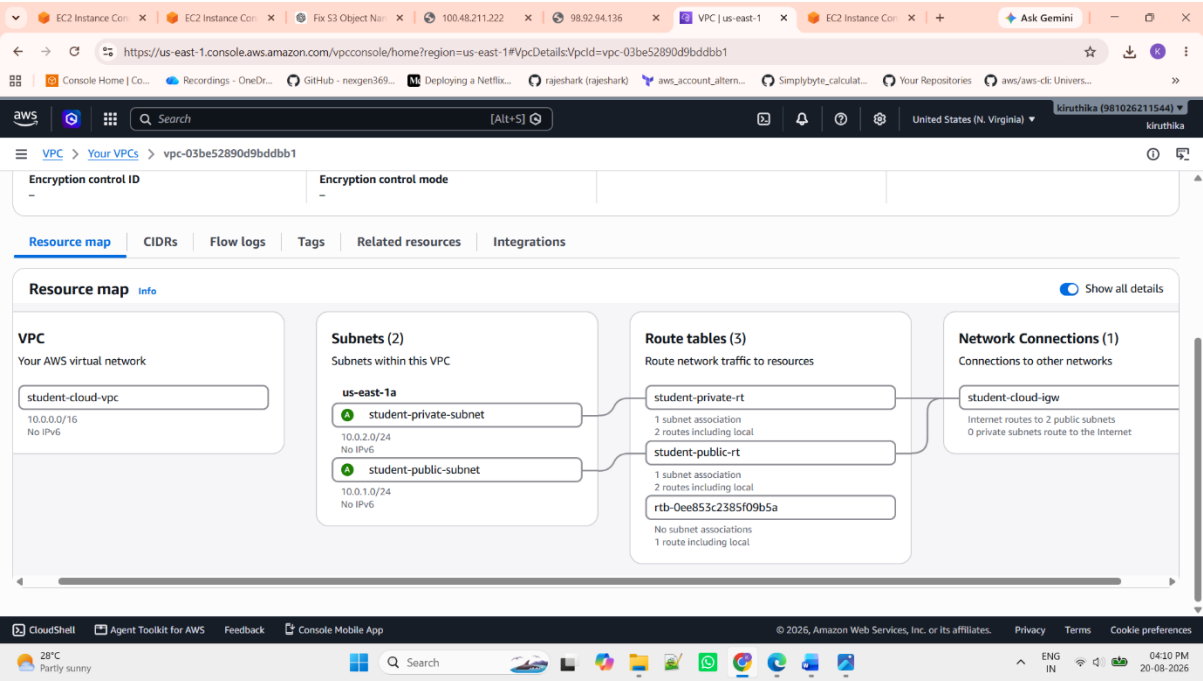
IAM Role:



The screenshot displays the AWS Management Console for the IAM role **StudentCloudEC2Role**. The role is attached to the EC2 instance **i-0a07daa5308b61e77**. The role has the **AmazonEC2RoleForLinux** policy attached. The console shows the role's name, ARN, and the list of attached policies.

Field	Value
Role name	StudentCloudEC2Role
Role ARN	arn:aws:iam::981026211544:role/StudentCloudEC2Role
Attached policies	AmazonEC2RoleForLinux

Task4: Networking



Bonus:

Browser showing Hello from student-public-ec2:

